# Econ 812 Recitation 8 Notes

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## 1 Notes on CE

In the competitive equilibrium, introduce prices and the concept of the firm and households.

The firm's problem is to maximize profit:

$$\max_{k_t, l_t} F(k_t, l_t) - r_t k_t - w_t l_t \tag{1}$$

taking FOC's of this problem leads to the market prices:

$$F_l(k_t, l_t) = w_t, F_k(k_t, l_t) = r_t$$

Additionally, households problem is similar to the SPP but they face a budget constraint rather than a resource constraint:

$$\max_{c_t, i_t} \sum_{t=0}^{\infty} \beta^t [u(c_t, l_t) + \lambda_t(-c_t - i_t + w_t l_t + r_t k_t + \pi_t)] \tag{2}$$

Where $i_t = k_{t+1}$ or $i_t = k_{t+1} - (1-\delta)k_t$

Then how you define a CE is the following: "a sequence of prices $P = \{w_t, r_t\}$, plans for the HH $H = \{c_t, k_t, i_t, l_t\}$, plans for the firm $F = \{k_t, l_t, \pi_t\}$ subject to given prices and an initial amount of capital, H solves the HH problem, F solves the firms problem, and markets clear $y_t = c_t + i_t$, $k_t^s = k_t^d$ and $l_t^s = l_t^d$"

## 2 Final 2022 Question 2

H.H. have the following preferences:

$$E_0[\sum_{t=0}^{\infty} \beta^t [\log(c_t) - 1/2 n_t^2]]$$

with $0 < \beta < 1$. The HH supplies capital services $z_t := u_t K_t$ and labor $n_t$ in competitive markets. $K_t$ is the capital stock the HH owns and $u_t$ is the utilization rate (between 0 and 1) of that capital stock. The HH BC is

$$c_t + I_t = r_t z_t + w_t n_t + \pi_t$$

and the evolution of capital is given by $K_{t+1} = [1 - u_t^2]K_t + I_t$. The HH selects $c_t, I_t, n_t$ and $u_t$ to maximize utility subject to the constraints. The firm has technology

$$Y_t = A_t Z_t^{\alpha} N_t^{1-\alpha}$$

Where TFP follows $\log A_{t+1} = 0.9 \log A_t + e_{t+1}$ and the firm selects $Z_t$ and $N_t$ to max profits taking $r_t, w_t, A_t$ as given.

Let $\alpha = 1/2$ and $\beta = 0.6$

## 2.1 Part A

Setup the HH lagrangian and derive FOCs and TVC [10 pts]

$$y_t = A_t z_t^{\alpha} N_t^{1-\alpha}$$

$$z_t = u_t k_t$$

Set up lagrangian:

$$\max_{c_t, n_t, u_t, k_{t+1}} \mathscr{L} = E_0[\sum_{t=0}^{\infty} \beta^t [\log C_t - 1/2 N_t^2 + \lambda_t [r_t u_t k_t + (1 - u_t^2) k_t + w_t n_t + \pi_t - c_t - k_{t+1}]]]$$

Take FOC's

$$c_t : 1/c_t = \lambda_t$$

$$n_t : n_t = \lambda_t w_t$$

$$u_t : r_t k_t = 2 u_t k_t \implies r_t = 2 u_t$$

$$k_{t+1} : \lambda_t = E_t[\beta \lambda_{t+1} [u_{t+1} r_{t+1} + 1 - u_{t+1}^2]]$$

Plus BC and TVC $= \lim_{j \to \infty} E_t[\beta^{t+j} \lambda_{t+j} k_{t+j+1}]$

## 2.2 Part B

Setup the firm's problem and derive FOC for profit max [5pts] FOC's

$$w_t = (1 - \alpha) A_t (z_t / N_t)^{\alpha}$$

$$r_t = \alpha A_t (z_t / N_t)^{\alpha - 1}$$

### 2.3 Part C

Solve for steady state. [10 pts]

1. $\log A = 0.9 \log A \implies A = 1$
2. $C + I = AZ^{\alpha} N^{1-\alpha}$
3. $1 = \beta[u\alpha A(Z/N)^{\alpha-1} + 1 - u^2]$
4. $I = u^2 K$
5. $CN = (1-\alpha)A(Z/N)^{\alpha}$
6. $Z = uK$
7. $r = \alpha A(Z/N)^{\alpha-1}$
8. $w = (1-\alpha)A(Z/N)^{\alpha}$
9. $r = 2u$

Here there are 9 equations with 9 unknowns, can substitute and solve from here. (Start with 3,7,9 to solve for $u$, then you immediately have $r$, then solve for $z/n$, then solve for $w$ and so on.)

## 3 Growth: Comp problem June 23

There is a continuum of households with representative household preferences over consumption given in equation 1.

$$\sum_{t=0}^{\infty} \beta^t \ln C_t \tag{3}$$

Where $C_t$ is consumption. The household has 1 unit of time that it supplies inelastically in the labor market. Capital evolves according to

$$K_{t+1} = (1-\delta)K_t + I_t$$

with the depreciation rate $\delta \in (0, 1]$ and $K_0 > 0$ given. The household's budget constraint is

$$C_t + I_t = r_t K_t + w_t + \pi_t$$

where $\pi_t$ is any profit earned by the firms - taken as given by the households. There is a unit continuum of firms. The representative firm has a production function given by

$$y_t = \Gamma A_t k_t^{1/2} n_t^{1/2}$$

with $\Gamma > 0$. The firm takes $A_t$ as given. The TFP term $A_t$ depends on the aggregate output last period $A_t = Y_{t-1}^{1/2}$. Market clearing in the goods market requires $C_t + I_t = Y_t$ .

## 3.1 Part A

Define a symmetric competitive equilibrium for this economy. [5 pts]

A symmetric competitive equilibrium is a set of prices $P = \{w_t, r_t\}$, a plan for the firm $F = \{k_t, n_t, \pi_t\}$ a plan for the households $H = \{K_t, I_t, C_t, N_t\}$ subject to given prices and an initial amount of capital, H solves the household's problem, F solves the firm's problem, goods markets clear $Y_t = C_t + I_t$, capital markets clear $k_t = K_t$ and labor markets clear $l_t = L_t$

## 3.2 Part B

Under what conditions on model parameters do we have a balanced growth path with positive growth. Under these conditions, solve for the balanced growth path. At what rates do consumption, wages, and output grow along the balanced growth path? [20 pts]

Set up the household's problem:

$$\max_{C_t, K_{t+1}} = \sum_{t=0}^{\infty} \beta^t [\ln C_t + \lambda_t [(r_t + 1 - \delta) K_t + w_t - C_t - K_{t+1}]]$$

Take FOC's:

$$C_t : \lambda_t = 1/C_t$$

$$K_{t+1} : \lambda_t = \beta \lambda_{t+1} [r_{t+1} + 1 - \delta]$$

Plus the BC and TVC.

Find the EE to be

$$1/C_t = \beta 1/C_{t+1}(r_{t+1} + 1 - \delta)$$

Since we have $A_t = Y_{t-1}^{1/2}$, we know that $\gamma_A = \gamma_y^{1/2}$ and from the definition of $y_t$ we have $\gamma_y = \gamma_A \gamma_k^{1/2}$. Substituting here find that $\gamma_y = \gamma_k$. Then from the BC and TVC, we know that $\gamma_c = \gamma_k$.

From the firm's problem we have $r_t = 1/2\Gamma A_t k_t^{-1/2}$, and from the definition of $\gamma_A$ we have

$$\gamma_A = \frac{A_{t+1}}{A_t} = \frac{Y_t^{1/2}}{A_t} = \frac{(\Gamma A_t k_t^{1/2})^{1/2}}{A_t} = (\frac{\Gamma k_t^{1/2}}{A_t})^{1/2}$$

So combining with the Euler Equation:

$$\gamma_c = \beta(1/2\Gamma A_t k_t^{-1/2} + 1 - \delta) = \frac{\Gamma k_t^{1/2}}{A_t}$$

Solving for $A_t k_t^{1/2}$ we find

$$1/2\Gamma (A_t k_t^{-1/2})^2 + A_t k_t^{-1/2}(1 - \delta) - \frac{1}{\beta}\Gamma = 0$$

Solving using the quadratic formula we have that

$$A_t k_t^{-1/2} = \frac{-(1-\delta) \pm \sqrt{(1-\delta)^2 + 2\Gamma^2/\beta}}{\Gamma}$$

Take the positive solution here and confirm that the ratio of $\frac{A_t}{k_t^{1/2}}$ is constant. Can plug this solution back into the growth rate expressions to solve for each.

### 3.3 Part C

How does an increase in $\delta$ affect the growth rate and level of output in this economy? Explain. How does an increase in $\Gamma$ affect the growth rate and level of output in this economy? Explain. [16 pts]

Take the derivative of $A_t k_t^{-1/2}$ w.r.t $\delta$ to find

$$\frac{1}{\Gamma} + \frac{1/2((1-\delta)^2 + 2\Gamma^2/\beta)^{-1/2}}{\Gamma} * 2(1-\delta) > 0$$

So then if $\delta$ increases then $A_t k_t^{1/2}$ increases then $k_t^{1/2} A_t^{-1}$ decreases then $\gamma_A$ decreases and the level of output decreases. This makes sense since if capital is depreciating faster then the economy stagnates.

For $\Gamma$, consider that $\gamma_y = \Gamma \frac{k_t^{1/2}}{A_t}$ so an increase in $\Gamma$ would increase $\gamma_y$ directly, but it would also affect $\frac{k_t^{1/2}}{A_t}$. In the limit, we can see that an increase in $\Gamma$ would leave the ratio unchanged, and so we can say that increasing technology causes output to grow and the rate of growth to be higher. Again this makes sense since better technology leads to more output (like how migrating from pen and paper to microsoft excel would increase productivity).

### 3.4 Part D

Is the competitive equilibrium Pareto optimal? Explain. [9 points]

It would be Pareto optimal if the SPP leads to the same outcome. In that case

$$\max_{C_t, K_{t+1}} \sum_{t=0}^{\infty} \beta^t [\ln C_t + \lambda_t [F(K_t, N_t) - C_t - K_{t+1} + (1-\delta)K_t]]$$

FOC's:

$$C_t : \lambda_t = 1/C_t$$

$$K_{t+1} : \lambda_t = \beta \lambda_{t+1} [\Gamma 1/2 A_{t+1} K_{t+1}^{-1/2} + 1 - \delta]$$

This is the same EE as above, so yes the CE is Pareto Optimal.